Broking Revenue & Client Intelligence Analytics

Project Documentation | Power BI · DAX · Power Query · Python

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Independent portfolio project built on a fully synthetic dataset. It models common patterns of Indian discount broking platforms and is not affiliated with any company.

1. Business Problem & Structural Risk

Discount broking platforms charge zero brokerage on equity delivery and mutual funds, and a flat fee per order on intraday and derivatives. Account opening is free. This model is very good at acquiring clients, but it carries three structural risks that standard reports do not show:

- **Dormant accounts.** Free onboarding brings clients who trade once or twice and go quiet. Each one costs money to acquire and verify, but earns almost nothing because their orders are mostly zero-brokerage products.
- **Revenue concentration.** Delivery and mutual funds earn ₹0, so derivatives (F&O) carry the business. A slowdown in F&O activity hits revenue directly.
- **Reliability at peak load.** Order volume peaks at market open and on F&O expiry days. A failed order at that moment is lost brokerage and lost client trust.

This project answers two questions:

- Which client segments and products drive durable revenue, and how much revenue do dormant clients actually contribute?
- When, why and how often do orders fail, and how does trading behaviour change on expiry days?

2. Data Pipeline & Metrics Engineered

2.1 Dataset (synthetic)

All data is synthetic, generated with Python (Pandas, NumPy, Faker), so no real client or market data and no personal information is involved. It covers Jan 2024 to Jun 2025.

Table	Rows	What it holds
Dim_Customer	10,000	City, city tier, acquisition channel, age band, KYC status
Dim_Product	15	7 segments: Equity Delivery, Intraday, F&O Futures, F&O Options, Commodity, Currency, Mutual Fund
Date Table	547 days	Calendar with weekend, market-day and expiry-day (Tue/Thu) flags
Fact_Trades	208,980 (210,024 raw)	One row per order: client, product, date, time, value, brokerage, status, failure reason

Behaviour was built into the data so the analysis has real patterns to find: flat per-order brokerage (₹20 for intraday, F&O, commodity and currency; ₹0 for delivery and mutual funds), heavier F&O trading on expiry days, a higher failure chance in the 9:15–10:00 AM window, and different client types (one-time traders who go dormant, casual investors, regular traders and heavy F&O traders).

2.2 Processing the raw trade log

Step	What was done
Clean (Power Query)	Profiled and counted 1,044 duplicate trade IDs (0.5%), then removed them (210,024 to 208,980 rows). Fixed city names with wrong case and stray spaces (about 17.5% of clients) using Trim and Proper; distinct cities collapsed to the true 16. Validated date, decimal and boolean types.
Model	Star schema: Dim_Customer, Dim_Product and Date Table each connect one-to-many to Fact_Trades with single-direction filtering. Foreign keys hidden in report view. The date table was generated in Python, imported, and marked as the model's Date Table.

Step	What was done
Engineered columns (DAX)	**Trade Hour** from trade time, to find the peak-hour pattern. **Days Since Last Trade** per client (latest trade per client versus the latest date in the data, using ALLEXCEPT, ALL and DATEDIFF). **Client Status**: Dormant if 60 or more days since the last trade, otherwise Active.
Flags	**is_expiry_day** (Tuesday/Thursday) and **is_market_day** in the Date Table, so expiry behaviour can be isolated in DAX.

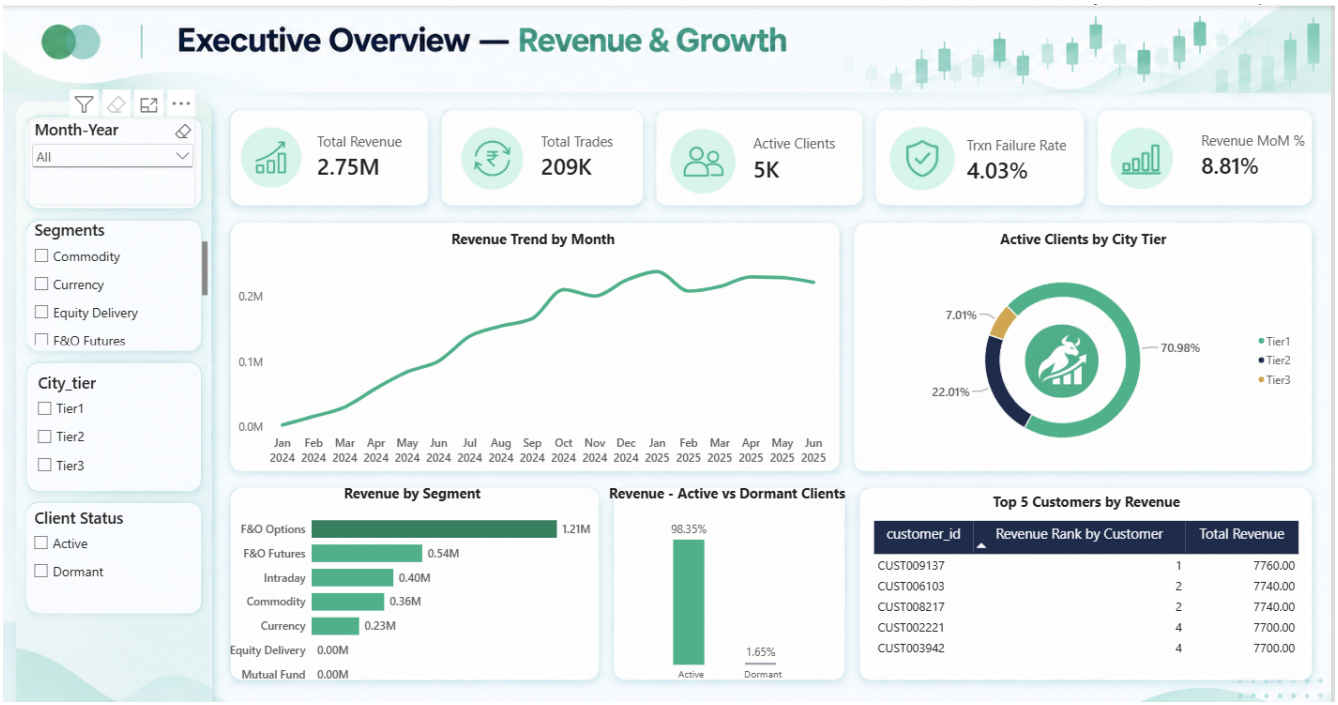
2.3 Metrics engineered

Metric	Logic	Value
Total Revenue / Total Trades	Sum of brokerage on successful orders / order count	₹2.75M / 209K
Active Clients	Distinct clients with Active status	5K
Transaction Failure Rate	Failed orders ÷ all orders	4.03%
Dormancy Rate	Clients with 60+ days since last trade ÷ all clients	49.38%
Revenue per Active Client	Revenue from active clients ÷ active clients	₹533.51
Peak-Hour Failure Rate	Failure rate for orders between 9:15 and 10:00 AM	8.87%
Expiry Day Volume Multiplier	Avg F&O Options trades per expiry day ÷ per normal trading day	2.09x
Revenue MoM %	Month-over-month revenue growth using the Date Table	8.81%

Validation. Every headline number was checked independently against the source data. One error was caught and fixed: the first version of the Expiry Day Multiplier showed 3.47x because weekends were counted as non-expiry days. Dividing by trading days only gives the correct 2.09x.

3. Key Insights & Financial Impact

3.1 Executive view: where the revenue comes from



Executive Overview page: KPIs, monthly revenue trend, revenue by segment, active vs dormant revenue and top 5 clients.

Insight	Number
Revenue is concentrated in derivatives. F&O Options (₹1.21M) and F&O Futures (₹0.54M) together earn about 64% of ₹2.75M.	₹1.75M of ₹2.75M
Delivery and mutual funds earn nothing by design, so they bring clients and volume but no brokerage.	₹0 brokerage
Dormant clients are half the base but almost no revenue. A dormant client earns about ₹9 on average, an active client ₹533.51, roughly 58 times more.	49.38% of clients, 1.65% of revenue

3.2 Risk view: where revenue leaks

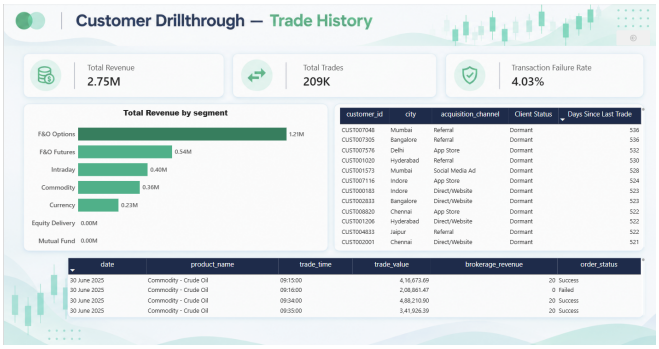


Client & Market Risk Intelligence page: dormancy, channel quality, failure reasons, failure by hour and expiry-day activity.

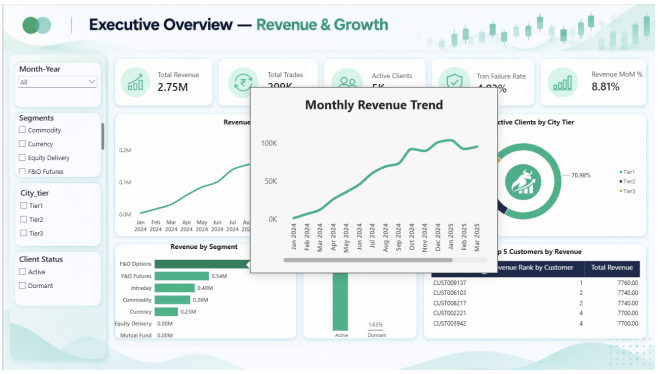
Insight	Number
Failures spike at market open. The 9:15–10:00 window fails about nine times as often as the rest of the day (about 1%).	8.87% vs 4.03% overall
Server timeouts are the biggest cause of failed orders, followed by margin shortfall and exchange connectivity.	4.2K / 3.0K / 1.2K of 8.4K (50% / 36% / 14%)
Expiry days are far busier. F&O Options trades per day on expiry days versus normal trading days.	234 vs 112 = 2.09x
Channel quality differs. App Store clients earn the least per active client; Referral the most.	₹496.73 vs ₹562.62 (about 12% lower)

3.3 Drilling into one client or one segment

Two interactive features let an analyst move from the summary to the detail without leaving the report.



Customer Drillthrough: right-click a client in the Top 5 table to see that client's profile and full trade history (shown here before a client is selected).



Segment tooltip: hover over a segment bar to see that segment's monthly revenue trend.

4. Product Recommendations & Risk Mitigation

Each recommendation is tied to an insight above, an owner, and a metric that shows whether it worked.

#	Insight	Recommended action	Owner	Success metric
1	Peak-hour failures at 8.87%; server timeouts are half of all failures	Load-test and add capacity for the 9:15–10:00 AM window; add queueing and safe retry for orders during the open	Engineering	Peak-hour failure rate moves toward the all-day average
2	Margin shortfall causes 36% of failures	Show a margin check or warning in the order window before the order is sent	Product	Share of margin-shortfall rejections
3	49.38% of clients are dormant and earn 1.65% of revenue	Trigger a re-engagement journey before a client reaches 60 days of inactivity, and nudge a first paid product (for example intraday), since delivery earns ₹0	Marketing, Product	Dormancy rate; number of reactivated clients
4	Expiry-day volume is 2.09x normal for F&O Options	Plan capacity and monitoring for Tuesdays and Thursdays, and track failures separately on expiry days	Engineering, Risk	Failure rate on expiry days
5	App Store clients earn about 12% less per active client than Referral	Review App Store targeting and onboarding; consider shifting acquisition budget toward referral	Marketing	Revenue per active client by channel

Possible extensions

- Estimate the brokerage lost to failed orders, to put a rupee value on recommendation 1.
- Add cohort retention (how many clients from each signup month are still active after 1, 2 and 3 months).

Assumptions and limits

All figures come from synthetic data, so they demonstrate the analysis method and are not claims about any real market or company. Brokerage is a simplified flat fee per order, dormancy uses a 60-day threshold, and expiry days are simplified to Tuesday and Thursday. Failed orders are assumed not to be retried. The recommendations show how the insights would be turned into actions by a product or risk team.